

AIDAN MCCARTHY

Senior Research Engineer

Email: aidan.mccarthy@gmail.com

Phone: +353 87 456 2891

LinkedIn: [linkedin.com/in/aidanmccarthy](https://www.linkedin.com/in/aidanmccarthy)

Location: Dublin, Ireland

PROFESSIONAL SUMMARY

Results-driven Research Engineer with 12 years of experience in advanced systems development and machine learning infrastructure. Proven track record of architecting large-scale distributed systems and leading cross-functional teams through complex technical challenges. Passionate about pushing the boundaries of what's possible in research computing and delivering innovative solutions that drive competitive advantage. Strong background in cloud technologies, data engineering, and emerging AI frameworks.

WORK EXPERIENCE

Research Engineer

TechVision Labs

www.techvisionlabs.io

Dublin, Ireland

June 2022 - Present

Directed and managed a team of 35 research engineers and data scientists across three international offices, overseeing all technical decisions and strategic roadmap planning for the entire R&D division.

Architected and deployed a proprietary machine learning framework that processes 2 petabytes of data daily, resulting in 40 percent improvement in inference latency across production systems.

Led the design and implementation of a distributed computing cluster spanning 500+ nodes, enabling real-time anomaly detection for 150 major financial institutions.

Established and enforced engineering standards across the organization, reducing critical production incidents by 65 percent through comprehensive code review automation and testing protocols.

Junior Systems Engineer

DataFlow Solutions

www.dataflowsolutions.net

Cork, Ireland

March 2021 - May 2022

Managed a cross-functional team of 28 backend engineers responsible for maintaining the core infrastructure serving 2 million concurrent users globally.

Developed and maintained internal Python libraries for data processing pipelines, achieving 98.7 percent uptime across all services handling real-time analytics.

Single-handedly designed the company's disaster recovery strategy and implemented automated failover mechanisms across multiple cloud regions.

Conducted comprehensive performance optimization analysis reducing database query times by 72 percent, saving the company an estimated 1.2 million euros annually in cloud infrastructure costs.

Graduate Software Developer

Insight Analytics Group

www.insightanalytics.co

Dublin, Ireland

July 2012 - February 2021

Completed graduate rotation program and progressed through multiple technical roles over nine years with increasing responsibility and scope.

Implemented full-stack web applications using React and Node.js, contributing to features used by over 500,000 active users in production environments.

Collaborated with senior architects on foundational systems design, participating in decisions that shaped the company's technical direction for machine learning initiatives.

Mentored three junior developers and contributed to the establishment of internal best practices documentation and coding standards.

EDUCATION

Bachelor of Science in Computer Science

University College Dublin

Dublin, Ireland

Graduated: May 2012

Grade: Upper Second Class Honours

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, Java, C++, Go, SQL, R

Cloud Platforms: Amazon Web Services, Google Cloud Platform, Microsoft Azure

Big Data Technologies: Apache Spark, Hadoop, Kafka, Elasticsearch

Machine Learning: TensorFlow, PyTorch, Scikit-learn, XGBoost

Databases: PostgreSQL, MongoDB, Redis, Cassandra, DynamoDB

DevOps and Tools: Docker, Kubernetes, Jenkins, GitLab CI/CD, Terraform

Distributed Systems: Microservices Architecture, Service Mesh, Load Balancing

Data Engineering: ETL Pipelines, Data Warehousing, Stream Processing

Version Control: Git, GitHub, GitLab

Methodologies: Agile, Scrum, Kanban